

SOWMIYA D

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PROFESSIONAL SUMMARY

Computer Science graduate (2026) with hands-on software engineering experience building production-quality software in collaborative engineering environments. Strong problem-solving skills with a passion for designing scalable applications, learning new technologies, and contributing to high-impact engineering teams.

EDUCATION

B. E. Computer Science and Engineering	Sep 2022 – Jun 2026
<i>College of Engineering Guindy, Anna University</i>	8.17/10
12th Grade	Jun 2021 – Jun 2022
<i>Bharathidasan Matric Higher Secondary School, Tiruvallur</i>	97.33%
10th Grade	Jun 2019 – Jun 2020
<i>Bharathidasan Matric Higher Secondary School, Tiruvallur</i>	91.20%

SKILLS

Programming Languages: C++, C, Java, Python, SQL
Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems (DBMS), Computer Networks
Backend Development: Flask, REST APIs, Redis, MySQL, Linux, IPC (Inter-Process Communication)
Cloud & DevOps: AWS (EC2, S3, IAM), Azure (Fundamentals), Docker, Kubernetes, Terraform, GitHub Actions, CI/CD
AI / Machine Learning: TensorFlow, Keras, PyTorch, scikit-learn, Pandas, NumPy, RAG, FAISS, ChromaDB
Tools & Build Systems: Git, GitHub, Linux Shell, CMake, Make, GCC, Yocto, Brazil

PROFESSIONAL EXPERIENCE

Amazon – Software Development Engineer Intern	Jan 2026 – Jun 2026
<i>Kindle Platform Team C, C++, Linux, Git, CMake, Brazil, Yocto Certification</i>	<i>Chennai</i>
- Developed and integrated a BLE-based Wi-Fi provisioning framework to replace the legacy device setup stack on Kindle E-ink devices, enabling secure production device registration and improving the device onboarding experience. - Designed and implemented a production-grade C++ daemon with 3 Hardware Abstraction Layer (HAL) interfaces for Wi-Fi and device information, adding WPA3 support and improving provisioning reliability across multiple Kindle hardware variants. - Redesigned inter-process communication (IPC) between networking services to resolve integration bottlenecks, implemented automatic device state transitions, and collaborated through Git-based development workflows with 20+ peer code reviews, contributing 7 production code changes that were successfully reviewed and merged. - Authored 10 technical design and implementation documents covering system architecture, feature design, setup workflows, and implementation details, improving knowledge sharing, maintainability, and onboarding for future development. - Debugged, validated, and verified provisioning workflows across 20+ Kindle hardware variants during a platform software migration, ensuring 100% successful device registration and backward compatibility within an Agile software development environment.	

PROJECTS

TaskFlow – Distributed Task Processing System GitHub	Jul 2026 – Present
Designed and developed a distributed task processing system with scalable API and worker services using Redis-based job queues for asynchronous execution, containerized with Docker, orchestrated using Kubernetes, and provisioned AWS (S3, IAM) infrastructure with Terraform for reproducible deployments.	
Marginalia – AI Document Assistant GitHub	Jun 2026 – Jul 2026
Built a Flask-based AI document assistant using Retrieval-Augmented Generation (RAG) with FAISS/ChromaDB for document ingestion, semantic search, and question answering, deployed on AWS EC2 with Docker and automated using GitHub Actions CI/CD.	
Lumora – AI Mental Healthcare Chatbot GitHub	Jan 2025 – May 2025
Developed an NLP-based mental healthcare chatbot using a 3-layer TensorFlow neural network for intent classification and context-aware conversations, implementing text preprocessing, feature extraction, and response generation for improved user interaction.	

LEADERSHIP

Head – Computer Society of Anna University (CSAU)	May 2025 – Jun 2026
<i>Led a 20-member Marketing & IR vertical, landing a Rs. 1,00,000 title sponsorship from Hay Clothing.</i>	<i>Anna University</i>
Associate Head – Anna University Student Entrepreneurship Club (AUSEC)	Mar 2024 – Mar 2025
<i>Managed sponsorship and industry relations for Enantra 9.0, securing Rs. 35,000 from Trends Clothing.</i>	<i>Anna University</i>